The Local Economic Impact of Data Centers

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July 13, 2026

Abstract

A hyperscale data center costs over a billion dollars but employs only dozens of people. We ask whether these investments generate the local agglomeration spillovers that large plant openings have historically produced. Comparing isolated built facilities to announced-then-cancelled sites, we find a large but sharply localized capital footprint in nighttime lights that fades within five kilometers. Advertised salaries and new-firm formation do not rise at opening. Hiring and supplier-posting estimates are also centered near zero, although they are less precise, and detailed county-industry data are too suppressed and trend-contaminated to identify downstream effects cleanly. Rents are modestly higher in synthetic comparisons, but imprecise relative to cancelled sites, and net migration does not rise. School property-tax revenue rises suggestively, while capital outlays are positive but imprecise. Residential electricity rates show no increase and suggestive declines. The evidence points to a large physical investment with limited measurable local propagation.

Keywords: data centers; place-based investment; local labor markets; nighttime lights; agglomeration.

JEL Codes: R11, R23, O33, H71.

1 Introduction

The physical infrastructure of digital services has become a major component of US capital investment. Data centers consumed roughly four percent of US electricity in 2023, and the rise of artificial intelligence has sharply accelerated new construction. A single hyperscale campus typically costs five hundred million to two billion dollars and draws one hundred to five hundred megawatts of continuous power, yet employs only a few dozen people on site. At least thirty-five states now offer tax incentives aimed specifically at data centers, with cumulative subsidies approaching twenty billion dollars.[1] Those subsidies are justified by the promise of local economic gains. This paper asks whether the gains materialize.

The natural benchmark is the canonical result on large plant openings. Greenstone et al. [2010] identify these spillovers by comparing counties that won a large new plant to the runner-up counties that lost it, and show that the productivity of incumbent establishments in winning counties rises by about twelve percent over the following five years. The gain operates through three agglomeration channels, namely a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. This result is the empirical backbone of the case for place-based subsidies. At the same time, subsequent work contests this result, finding that estimates from the winner-versus-loser design are sensitive to the choice of comparison counties [Patrick, 2016], and the average large plant shows little evidence of productivity spillovers [Patrick and Partridge, 2022]. The spillover evidence is therefore contested even for manufacturing plants, the historically strongest case for agglomeration.

A data center is large in terms of investment dollars, but it employs dozens of people, not thousands, and so does not deepen the local labor pool. Moreover, its main inputs (electricity and fiber-optic connectivity) are purchased through national markets and its output (compute) is delivered over the cloud to customers around the world. Given these structural differences from manufacturing plants, it is an open question whether the canonical agglomeration result extends to the signature capital investment of the AI era. The subsidies at stake make the question a consequential one.

Operators screen sites on cheap power, grid capacity, land cost, and fiber, so a county that hosts a data center differs from one that does not in ways that also shape the local economy. We document that this selection cannot be easily instrumented away. All of the predetermined features that we test as predictors of hyperscale siting are also correlated with local industrial character, and the features that are clean do not predict where these facilities go (Section 3).

Our primary design is therefore the analog of the winner-versus-loser comparison Greenstone et al. [2010] used to identify plant spillovers. We compare built facilities to announced-then-cancelled ones, which were sited for similar reasons but never built. The cancelled cohort contains eighty-four candidate projects, of which fifty-two enter the nighttime-lights design. We also report two supporting designs. First, we apply a within-site ring comparison, comparing each facility's

[1] Authors' tally from the Good Jobs First Subsidy Tracker, which records state and local data-center subsidy awards and is one of the sources for our facility registry (Section 2).

inner rings to its own twenty-five to fifty kilometer ring, which provides estimates for outcomes that pass a pre-trend test. Second, we use county panels to test whether aggregate industry outcomes change at opening, while treating their suppression and pre-trends as limits on what they can establish.

We first establish that the capital footprint is real, large, and local. Relative to cancelled sites, nighttime lights at isolated built sites jump at the start of construction within the first kilometer and decay to near zero by five kilometers. Daytime imagery confirms that vegetation clears when the lights appear. We then turn to the evidence for agglomeration, testing each of the three mechanisms. Hiring in the surrounding rings shows no detectable rise, though the estimate is imprecise. Advertised salaries and total employment do not rise. There is no increase in new-firm formation.

The clearest local incidence is outside the labor market. Residential rents are two to five percent higher in synthetic comparisons, although the built-versus-cancelled estimate is imprecise, and multifamily permitting does not rise. Net migration is near zero. School property-tax revenue rises suggestively, while capital outlays are positive but imprecise and the county public-works results do not survive the staggered-treatment audit. Foot traffic and commercial spending show no robust change relative to cancelled sites. Residential electricity rates do not increase after a hyperscale opening and decline modestly in several specifications; the staggered-robust averages are less precise than the original event study, so we treat the fixed-cost interpretation as suggestive.

Taken together, the results show a large physical investment with little measurable propagation to nearby workers or firms. Data centers have recently been studied at the county level [Bahar and Wright, 2026, Alvarez et al., 2026].[2] Our within-site design adds the sub-county capital footprint, direct tests of the agglomeration channels, and evidence on local incidence. The contribution is to show where the response is visible and where the available designs do not detect one. This distinction matters as place-based investment shifts from labor-intensive plants toward labor-light digital capital.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 sets out the identification strategy. Section 4 documents the capital footprint. Section 5 tests the agglomeration channels and the county employment composition. Section 6 traces the local incidence to land. Section 7 examines retail electricity rates. Section 8 interprets the findings. Section 9 concludes. Appendices report the instrument search, the geography of compute, and supplementary results.

[2] Our own county-level analysis [Bahar and Wright, 2026] reported a positive downstream association using a synthetic-control design. Section 5.5 explains why the detailed county-industry data do not cleanly adjudicate that result: employment is heavily suppressed and several establishment series rise before opening. We therefore do not use those data as causal evidence for either a downstream gain or a precise zero.

2 Data

Data center registry. We build a master registry of US data centers from four sources. The base layer is the Open Source Data Center Atlas (Pacific Northwest National Laboratory's IM3 project), which identifies 1,242 facilities from OpenStreetMap footprints tagged as data centers, with coordinates and, where available, square footage and operator. We augment it with operational-date information from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, and trade press, and with four research batches targeting state announcements, specialty wholesale operators, AI-wave operators, and REIT filings. The final registry contains 1,494 facilities. The geocoded analysis backbone contains 1,118 facilities, including 341 hyperscale-tier facilities. Of these, 315 have high- or medium-confidence analysis dates inside the 2012–2024 VIIRS window. We assign hyperscale tier by operator identity (Amazon, Microsoft, Google, Meta, Apple, Oracle, plus hand-curated large specialty and AI-wave campuses). We do not impose a megawatt threshold, since capacity is not consistently observed. We also assemble the cancelled cohort for the built-versus-cancelled primary design. It contains eighty-four announced-then-cancelled projects, of which fifty-two have the coordinates and satellite coverage to enter the nighttime-lights design. Section 3 reports the cohort's sponsor composition and shows that the capital result is robust to restricting the control arm to projects with a named hyperscale tenant.

Nighttime lights. We extract monthly mean radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band Monthly Composite for April 2012 through December 2024, in concentric rings of 0–1, 1–2, 2–5, 5–10, 10–25, and 25–50 kilometers around each facility, aggregated to annual means. We classify each facility as isolated if no other data center within two kilometers opened earlier, and use the isolated, construction-dated sample for the headline capital-footprint estimate.

Local labor, housing, and spending panels. We measure local labor demand from the LinkUp Job Records panel [LinkUp, 2025] (postings by employer and geocoded location, by ring and quarter), residential rents from the RentHub Rental Data panel [RentHub, 2022] (median list price by ring and month), consumer spending from the SafeGraph Spend Patterns panel [SafeGraph, 2022], and foot traffic from the Advan Weekly Patterns panel [Advan Research, 2025], all distributed by Dewey Data. Each enters the within-site ring design dated to operations, with state-by-month fixed effects.

Realized county employment. For the county-level diagnostic we use industry employment and establishment counts from the Quarterly Census of Employment and Wages. We retain the Bureau of Labor Statistics disclosure flag, which is essential because suppressed employment is zero-filled in the public files even though the value is not observed. We focus on data processing (NAICS 518), telecommunications (517), web and information services (519), IT consulting (5415), and professional, scientific, and technical services (541). We end the 519 series in 2021 because the 2022 NAICS revision moved major components of that sector to 513 and 516. Section 5.5 reports both the disclosure rates and the pretrend diagnostics that limit causal interpretation of these county outcomes.

Instrument-search inputs. The instrument search draws on predetermined county features, namely 345 kV transmission-substation density, the InterTubes long-haul fiber network [Durairajan et al., 2015], the 1980 county share of urban college population (NHGIS), station-level climate normals (CustomWeather), and the EIA-860 generation fleet (Appendix A).

2.1 Dating construction and operations

We preserve announcement, groundbreaking, construction, planned-completion, and operational dates as separate events. Operator or official evidence of operation receives the highest weight. A lower-detail operational source receives medium confidence when its year agrees with an independent registry or satellite signal. Announcement-based projections, planned-completion dates, and unresolved source conflicts remain low confidence and are excluded from the clean-date analyses. This classification leaves 315 clean hyperscale dates and 26 low-confidence dates. For a smaller sample we recover construction timing from daytime satellite imagery. A hyperscale build begins with land clearing, which appears as a sustained drop in the parcel's vegetation index combined with a rise in its built-up surface index in the Landsat and Sentinel-2 scenes. Vegetation cover at the parcel is flat until the construction year and falls sharply after it. As an independent check, we join Federal Aviation Administration obstruction-evaluation filings to our projects. The isolated primary imagery sample contains 43 hyperscale facilities, while the independent FAA matches contain 64 facilities within 500 meters and 119 within one kilometer (Section 3.5). These independent dates validate the satellite construction timing used in the capital design.

3 Research Design

Data centers do not locate at random. Operators screen on cheap power, available grid capacity, large parcels of low-cost land, fiber proximity, and tax policy, so a county that hosts a data center differs from one that does not in ways that also move the local economy. A cross-county comparison would confound the data center with the conditions that attracted it. We document this selection directly. We attempted to instrument data center siting with a range of plausibly exogenous predetermined features, and each instrument failed, for reasons we report in Appendix A. The features that predict siting (power, industrial land, fiber) are themselves correlated with local industrial character, and the features that are clean (climate, historical human capital) do not predict where hyperscale facilities go.

We instead follow the design that anchors the agglomeration literature. Greenstone et al. [2010] identify plant spillovers by comparing counties that won a large plant to the runner-up counties that lost it, on the logic that winner and loser were selected on the same fundamentals. Our primary design is the data-center analog where we compare built facilities to announced-then-cancelled ones. Around it we use two supporting designs, each for what it does best. The within-site ring comparison supplies precision and spatial detail for outcomes that pass its pre-trend test. County panels provide an aggregate diagnostic, but the suppression and pre-trends documented below prevent us from treating them as a clean estimate of sectoral employment effects.

3.1 Built versus cancelled facilities

A facility that was announced and sited but never built is the loser to a built facility's winner. Built and cancelled sites were chosen for the same grid, land, fiber, and tax fundamentals, so a post-opening difference between them primarily reflects the fact that the facility was built, rather than the conditions that attracted it. We assemble eighty-four announced-then-cancelled hyperscale-scale projects from trade-press coverage, regulatory filings, and operator announcements, of which fifty-two enter the nighttime-lights design, and estimate, for each outcome,

$$y_{i,r,t} = \sum_k \left[\beta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r} \times \text{Built}_i + \delta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r}\right] + \alpha_{i,r} + \lambda_t + u_{i,r,t}, \quad (1)$$

where Built_i marks facilities actually built and t_i^c is the satellite-detected construction-start date for built sites and the announcement date for cancelled sites. The cancelled projects do not have an observed or consistently reported scheduled start, so their timing is not symmetric with the built arm. The path of β_k traces what building the facility does to its surroundings, net of anything common to places selected for one. For the capital footprint the comparison is especially sharp since a parcel that is never built never lights up, whatever the reason for cancellation.

The composition of the cancelled cohort deserves explicit note. Of the eighty-four candidate projects we track, fourteen had a named hyperscale operator publicly attached at cancellation, nineteen were sponsored by established data-center operators without a named tenant, and the remaining fifty-one, about sixty percent, were speculative developments by real-estate sponsors. A speculative loser is a weaker analog to the runner-up counties in Greenstone et al. [2010] than an operator-committed one, since it carries no signed tenant or confirmed grid allocation. Two facts limit the concern for our use of the design. For the capital footprint the comparison is mechanical, since a parcel that is never built never lights up whatever the sponsor. And the result does not depend on the speculative majority. Restricting the cancelled arm to the nine isolated sites where a hyperscale operator was named leaves the groundbreaking-year jump essentially unchanged, at seventy percent versus sixty-eight percent in the full isolated arm, though so thin a subset cannot support longer horizons. The cancelled sample is also somewhat small overall, and most cancellations followed local opposition, so cancelled sites may sit in somewhat less development-friendly places.[3] This is in part why we pair the design with the better-powered ring comparison

[3] Because most cancelled projects were announced late in the sample, the cancelled arm contributes few post-event observations. Coverage at event years zero through four is 51, 23, 6, 4, and 1 sites, so the built-versus-cancelled path at longer horizons is identified primarily from the built arm's post-opening change against the common pre-event trend rather than from post-period movement in the cancelled arm. Section 4 quotes magnitudes only through the

below. In addition, built sites are dated by satellite-detected land clearing, while cancelled sites carry scheduled dates from filings and trade press, which are softer. However, the independent regulatory-filing dates described in Section 3.5 validate the built-site dates as well.

3.2 Within-site rings

The cancelled comparison establishes whether an effect exists. It is too small to resolve how an effect decays over space or to deliver tight confidence intervals on nulls. For those we use the within-site ring design, which compares an inner ring of a facility to the same facility's twenty-five to fifty kilometer outer ring,

$$\log(1 + \mathrm{NTL}_{i,r,t}) = \sum_k \beta_k \, \mathbf{1}[t - t_i^c = k] \times \mathrm{Near}_{i,r} + \alpha_{i,r} + \gamma_{r,t} + \varepsilon_{i,r,t}, \tag{2}$$

where i indexes the facility, r the ring, t the year, t_i^c the facility's construction-start year, $\mathrm{Near}_{i,r}$ marks the inner ring, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and $\gamma_{r,t}$ is a ring-by-year fixed effect. The facility-by-ring fixed effect absorbs any time-invariant difference between a site's inner and outer ring, including the operator's siting preferences, the local industrial structure, and grid and fiber access. Identification comes from the change in the inner ring relative to the outer ring of the same site.

Two features of the design matter for credibility, and both correct a pre-trend present in a naive version of this specification. First, we date the event to construction start, not to the operational opening, to avoid confusing pre-period construction activity with operations (Section 2.1). Dated to construction, the inner-ring brightness is flat before groundbreaking and jumps at the groundbreaking date.

Second, we restrict the headline to isolated sites, meaning facilities with no other data center within two kilometers that opened earlier. A new facility built beside an existing one inherits the neighbor's brightness in its inner ring before it breaks ground, which could also generate a false pre-trend. The isolated, construction-dated event study has a flat pre-trend and then a jump of thirty-eight percent in the construction year that builds to a larger footprint over the following years. The jump survives controlling for a site-specific linear pre-trend estimated across the pre-construction years (the pre-trend is flat).

Importantly, we only report the ring estimates for outcomes whose pre-construction leads are flat. The capital footprint, rents, advertised wages, and firm entry pass this test. Posting counts and workplace employment do not, so for those outcomes the cancelled-project design carries our interpretation and we do not lean on the ring estimates in either direction. So that the reader can audit this routing rather than take it on faith, Appendix Table 6 reports every outcome under every design in which it can be estimated, with the estimate, its precision, and the reason any design is excluded for that outcome.

year after groundbreaking for this reason.

3.3 Demand-side outcomes

We use the same within-site ring comparison in equation (2) to identify three demand-side outcomes, namely residential rents, consumer spending, and foot traffic. We date these to the operational opening rather than to the construction start. The rent, spending, and foot-traffic panels are monthly and span the 2020 to 2023 period, so we replace the ring-by-year fixed effect with a state-by-month fixed effect, which absorbs the differential work-from-home and urban-recovery shocks across metropolitan areas in those years. The primary design identifies each of these outcomes, and the rings refine the spatial pattern. Relative to cancelled sites, rents rise modestly at built sites, foot traffic shows no robust change, and commercial spending is null in both arms. The rent, spending, and foot-traffic estimates enter the incidence analysis in Section 6.

3.4 Employment and county-industry diagnostics

Local labor demand is harder to identify locally relative to the capital footprint, so we measure it two ways. The first approach applies the within-site ring design of equation (2) to job postings near each facility, counting openings advertised within each ring.

The second approach is at the county level, because a data center's own employees and any firms that cluster around it need not fall inside a single ring. We use Quarterly Census of Employment and Wages employment and establishment counts by industry to estimate

$$\log(1 + \mathrm{Emp}^{s}_{c,t}) = \sum_{k} \theta^{s}_{k} \, \mathbf{1}[t - t^{o}_{c} = k] \times \mathrm{DC}_{c} + \mu_{c} + \lambda_{t} + u_{c,t}, \tag{3}$$

for industry s, county c, and the county's first data-center year t^{o}_{c}, with county and year fixed effects. We also estimate cohort-by-event-time effects using not-yet-treated counties as controls and state-cluster bootstrap inference. This audit changes the role of the county data. Between 27 and 68 percent of the detailed employment cells are suppressed, depending on sector, and establishment counts in several sectors rise before opening. We therefore use these outcomes to show why the county evidence cannot cleanly identify either a downstream gain or a precise zero. Section 5.5 reports the diagnostics and results in full.

3.5 Credibility of the capital-footprint design

The capital-footprint estimate is robust to the usual checks on a ring design. Dropping each facility in turn leaves the estimate in a narrow band, so no single site drives it, and a donut specification that uses the one-to-two kilometer ring as the inner contrast, omitting the campus pixel, returns the same spatial decay. Because most hyperscale sites sit near one another, we cluster standard errors at the campus, rather than the facility, level which widens the interval but leaves the effect statistically significant. Most important for construction timing, we re-date the facilities from Federal Aviation Administration (FAA) obstruction-evaluation filings, a regulatory timestamp recorded years before any radiance response and joined to a facility by location, with no satellite input. The construction-dated capital footprint is similar in sign and total magnitude under these independent dates. The FAA filing typically precedes physical construction, so this is a timing validation rather than an exact alternative groundbreaking date.

In Section 6 we perform one further check with respect to the rent result. Rents rose generally over 2020 to 2023 in the kinds of places that host data centers, and a within-site comparison would attribute that growth to the facility. We therefore build a synthetic control for each facility from population-matched locations with no data center, which nets out the common growth and isolates the data-center-specific rise.

4 The capital footprint

This section describes the capital footprint associated with data centers.

4.1 Built versus cancelled site design

Figure 1 reports the estimates from equation (1). This research design compares the inner-ring (zero to one kilometer) nighttime-lights path at built hyperscale sites to announced-then-cancelled sites. Built sites are dated to satellite-detected construction start; cancelled sites are dated to announcement because they do not have an observed construction start. We focus on *isolated* sites, which are the forty-three built facilities with no earlier data center within two kilometers of the parcel and the forty-nine cancelled sites away from operating campuses. The pre-construction coefficients are between eight and eleven percent below the omitted year. They are jointly indistinguishable from zero when clustered by campus ($p = 0.24$), and their linear slope is +3.5 log points per year ($p = 0.22$). Controlling for that slope leaves a sixty-nine percent jump in the construction year ($p < 0.01$). The event-study estimate is seventy percent and reaches roughly two hundred thirty-one percent above the cancelled-site benchmark a year later. The footprint includes activity at the parcel, its security and parking lighting, substations and cooling equipment, and continuous operation.

Two properties of this estimate matter for how it should be read. First, the cancelled cohort is concentrated in recent announcement years, so its coverage declines with the event horizon. Forty-eight control sites identify the groundbreaking-year contrast and twenty-one the year-one contrast, but five or fewer identify the contrasts beyond that. We therefore quote magnitudes through the year after groundbreaking, and we read the continued build-up at longer horizons, which reaches several hundred percent by year four, as suggestive rather than precisely identified. Second, the footprint belongs to the campus, not to a single building. Thirty-two of the forty-three isolated built sites add a sibling building within two kilometers during the event window, so the multi-year build-up reflects campuses phasing in additional buildings. Restricting to the eleven sites that stay single-building through year four, which are also the smallest projects, gives a footprint of about seven percent at groundbreaking, imprecisely estimated. We report the campus-level number as the headline because the campus is what a jurisdiction recruits. Clustering standard errors at the campus level widens the intervals by roughly sixty percent and leaves every post-groundbreaking coefficient statistically significant.

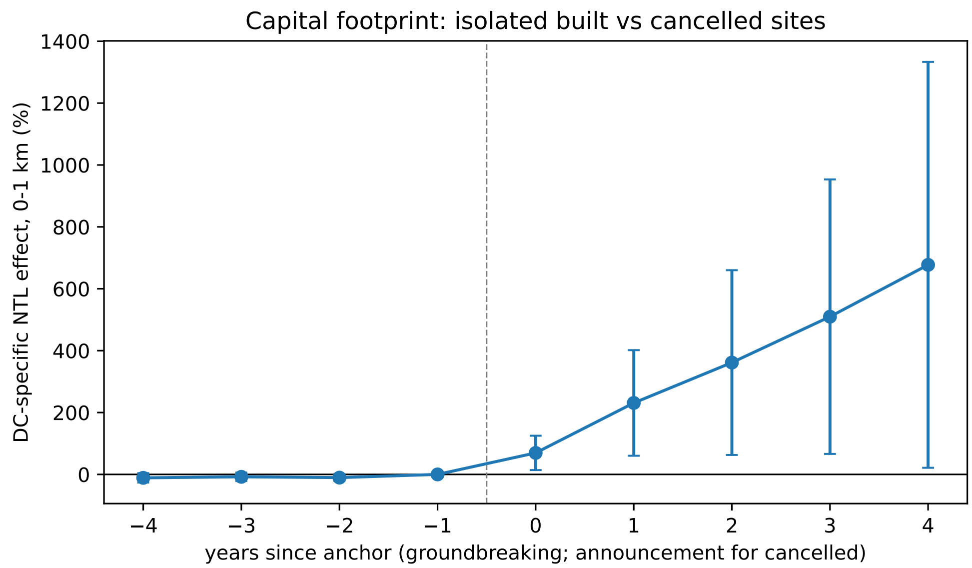

Figure 1: Capital footprint at built versus cancelled sites, dated to construction start

Notes: Event study of equation (1): the inner-ring (0–1 km) log nighttime-lights path at forty-three isolated built hyperscale facilities relative to forty-nine isolated announced-then-cancelled facilities, omitting the year before. Built facilities are dated by satellite-detected land clearing; cancelled facilities are dated by announcement year. Standard errors are clustered on two-kilometer campus groups. Cancelled-site coverage at years zero through four is 48, 21, 5, 3, and 1, so estimates beyond year one rest on few control sites.

4.2 Within-site ring design

The within-site ring design sharpens the estimate in two ways, each of which removes a source of contamination in the built-versus-cancelled design. The first is the dating. A hyperscale campus takes two to three years to build, so dating the event to the operational opening places the construction phase, when the parcel is physically changing, before the event window and reads it as a pre-trend. We date each facility instead to the construction-start year, recovered from satellite-detected land clearing (Section 2.1). Observing the construction start and the operational opening separately also prevents the construction period from being read as an opening-date pretrend.

The second issue is the sample. Roughly two in five hyperscale sites sit within two kilometers of an earlier data center, such that the site inherits its neighbor's brightness in its inner ring before it breaks ground. Splitting the sample makes this explicit. At sites with an earlier neighbor within two kilometers, inner-ring brightness rises about fifty percent in the years before construction. At isolated sites, where the inner ring is genuinely just the future parcel, the pre-construction excess is seven percent, and the event study is flat before groundbreaking. A joint test of the pre-construction coefficients does not reject ($p = 0.21$), and the jump is thirty-eight percent in the construction year, surviving a control for a site-specific linear trend estimated from the pre-construction years.

The two designs agree on timing and shape. Both show a flat pre-period at isolated sites followed by a sharp jump in the construction year and a multi-year build-up. The point magnitudes are not directly comparable because the two designs transform radiance differently, but each implies a construction-year brightening of several tens of percent at the parcel.

4.3 Spatial decay

The footprint is sharply localized. Estimated ring by ring, the effect is largest at zero to one kilometer, falls by more than half at one to two kilometers, and is small and fading by two to five kilometers, with no detectable effect beyond five kilometers. A donut specification that drops the zero-to-one kilometer ring and uses one-to-two as the inner contrast returns the same decay, so the result is not an artifact of the campus site itself. Overall, the site impact is large but it does not reach far.

4.4 Colocation as a within-design placebo

Colocation facilities in our sample are generally smaller, multi-tenant projects and often occupy or expand existing facilities, although some are purpose-built. They show no comparable changes in the capital footprint. Given this, the rest of the paper takes hyperscale facilities as the treatment whose spillovers, or absence of them, we measure.

5 Test of agglomeration

Section 4 establishes that a hyperscale data center is a large, spatially concentrated investment. The question is whether that capital generates the local spillovers that large plant openings have historically produced. Greenstone et al. [2010] trace those spillovers to three channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. We test each channel at the facility level, using the within-site ring design for outcomes that pass its pre-trend test and the built-versus-cancelled comparison for job postings, which do not. Colocation facilities provide a second comparison, with the caveat that they sit in denser markets than hyperscale campuses. We find no detectable opening-date response through these channels, although the hiring and supplier estimates are not precise enough to rule out moderate effects.

5.1 Labor pooling

Local labor pool depth should lead to more hiring and higher wages in the surrounding rings. We use the built-versus-cancelled design for job postings because the within-site series carries a differential inner-versus-outer trend. Relative to cancelled sites, postings near built facilities move by +4.0% ($p = 0.75$) with a flat pre-trend ($p = 0.96$). The estimate remains statistically indistinguishable from zero using Poisson counts, a long difference that controls for pre-period volume, and construction dating. This is not a precise zero. At conventional power the design could detect only a hiring increase of roughly forty percent or more, so it cannot rule out moderate gains.

Advertised salaries provide a narrower test. The mean log salary of nearby postings moves by -0.9% at 0–5 km ($p = 0.77$), with a ninety-five percent interval of $[-6.6, +5.2]$ percent. The panel is concentrated in salaried, posted positions, so the result is most relevant for white-collar workers and may not capture construction-trade compensation. The salary-disclosure rate is itself unchanged at opening ($p = 0.43$), so differential disclosure does not generate the result. Workplace employment from the Longitudinal Employer-Household Dynamics data carries a different limitation. Its raw within-site effect is $+8\%$ at 0–5 km, but a strong pre-existing trend explains the increase; detrending reduces the opening-date jump to -0.3%.

5.2 Supplier linkages

The second channel is local supplier linkages. We isolate postings in the sectors that build and equip a data center: construction and specialty trades, electrical-equipment wholesale, architectural and engineering services, and machinery manufacturing. The hyperscale supplier composite moves by $+1.7\%$ at 0–5 km ($p = 0.89$). Because supplier linkages may operate at the metropolitan level, we also compare the pooled 0–25 km zone to a 50–150 km control. That estimate is $+5.9\%$ ($p = 0.61$). The confidence intervals admit a moderate response, so the results do not establish a tight zero. They show that no large, precisely measured supplier cluster appears in the postings data.

5.3 Knowledge spillovers and firm entry

The third channel is knowledge spillovers among co-located firms. Appendix B maps heavy users of data-center capacity. They sit somewhat closer to data centers than the average firm, but much of this proximity reflects shared metropolitan location, and 69 percent of nearby compute firms pre-date the nearest facility. Exurban data centers often sit in places with few compute firms. These descriptive patterns do not identify an opening effect, but they give little indication of a cluster forming around the site.

New firm formation also shows no average jump at opening. We use Secretary-of-State new-registration records for Virginia and Texas geocoded to the facility rings. The within-site jump at 0–5 km is $+0.7\%$ ($p = 0.24$) in the pooled sample and -0.5% ($p = 0.62$) for hyperscale facilities. A national county design using Census Business Formation Statistics is also null ($+1.4\%$, $p = 0.49$). We therefore find no systematic entry response, not a uniform zero across locations.

5.4 Incumbent productivity

In Greenstone et al. [2010], incumbent establishments experience total factor productivity gains of around twelve percent. We do not observe plant-level productivity. We instead ask whether advertised salaries or hiring at incumbent firms rise, two outcomes through which a large local productivity gain could appear. The advertised-salary interval rules out a wage response as large as the twelve-percent benchmark if some of that gain reaches posted salaries. An event study of hiring by firms that existed before the facility shows no positive response, but the pooled and colocation samples carry a pre-existing downward trend that the design cannot fully remove. We therefore read incumbent hiring as corroborating rather than identifying and do not claim to observe productivity directly.

5.5 What county-industry data can establish

County-industry data are a natural way to look beyond job postings, but the detailed Quarterly Census of Employment and Wages cells have two limitations in this setting. First, the Bureau of Labor Statistics suppresses employment when publication could reveal an individual employer. The public files zero-fill these cells even though the employment level is not observed. Retaining the disclosure flag shows that suppression affects 68 percent of reported county-year records in data processing, 51 percent in telecommunications, and 66 percent in web and information services. Treating those values as observed zeros mechanically creates employment growth when a cell first becomes publishable.

Second, the host counties were already changing before the facility opened. We estimate cohort-by-event-time effects relative to the year before opening, using not-yet-treated counties as controls and a state-cluster bootstrap. Table 1 reports the average effect over event years zero through five and a joint test of the pre-period leads. Three of the ten hyperscale outcome series reject flat leads at the five-percent level, and five reject at the ten-percent level. This includes the data-processing establishment series, where the positive post-opening estimate is preceded by significant growth.

Table 1: County-industry disclosure and pretrend audit: hyperscale facilities

Industry (NAICS)	Employment suppressed	Employment effect	Pretrend p	Establishments effect	Pretrend p
Data processing (518)	68%	+5.2%	0.26	+27.2%***	0.13
Telecommunications (517)	51%	+5.2%	0.30	−2.6%	0.07
Web and information services (519)	66%	+9.3%	0.15	+18.3%***	0.04
Computer systems design (5415)	40%	−0.9%	0.24	+5.7%	0.06
Professional and technical services (541)	27%	+6.2%**	< 0.01	+7.6%***	< 0.01

Notes: Employment suppression is the share of reported county-sector-year records carrying the BLS non-disclosure code. Effects are cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated counties. Pretrend p is a joint test for event years −4 through −2. Inference uses 199 state-cluster bootstrap draws. The 519 series ends in 2021 because the 2022 NAICS revision moved major components to sectors 513 and 516. Stars refer to the post effect: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The county data therefore do not support the strong composition claim in earlier versions of the paper. Data-processing and telecommunications employment do not show a precise post-opening increase in the disclosure-corrected estimator. Several downstream outcomes are positive, but their pre-period leads reject the identifying assumption. Establishment counts avoid employment suppression, yet they reveal the same anticipatory growth rather than a discrete opening-date break. We consequently do not use this table as evidence that downstream growth occurs, or as evidence of a precisely estimated zero. Its contribution is diagnostic: county-industry trends cannot resolve the agglomeration question cleanly in this sample.

6 Incidence outside the labor market

The housing evidence is suggestive rather than definitive. The within-site rent estimates are +6.6% at 0–5 km, +9.4% at 5–10 km, and +8.5% at 10–25 km. A density-matched placebo shows that comparable non-data-center locations also experienced substantial rent growth during the 2019–2023 work-from-home period, so the within-site estimates are upper bounds. Facility-level synthetic controls return smaller estimates, between two and five percent across the three rings, although the procedure does not deliver formal placebo inference. The built-versus-cancelled estimate is +4.3% ($p = 0.23$), with a flat pre-trend ($p = 0.44$). We therefore read the evidence as a modest rent response, with considerable uncertainty about its size. Colocation rent estimates are null.

Neither housing supply nor net migration rises. A place-level Census Building Permits Survey check finds no multifamily permitting response, with permits for five or more units up 3.4% ($p = 0.87$). For migration, we assign each county the true first high- or medium-confidence hyperscale opening within 25 km, including openings before the IRS panel, and estimate cohort-by-event-time effects using not-yet-treated counties. Gross inflow and outflow series both have significant pre-trends, so we do not interpret them as opening effects. The log inflow-to-outflow ratio for returns moves by −1.4% over event years zero through five ($p = 0.52$), with a joint pretrend p-value of 0.14. The analogous adjusted-gross-income ratio moves by +0.4% ($p = 0.91$), with pretrend $p = 0.12$. The defensible conclusion is no evidence of net in-migration, not a causal increase in gross migration or migrant income.

Foot traffic and commercial spending also show no robust change relative to cancelled sites. The foot-traffic estimate is about −12% ($p = 0.12$) on the full panel but falls to −3% ($p = 0.68$) on the balanced panel. Commercial spending moves by +4.6% ($p = 0.60$). The within-site spending estimate is positive and passes a random-donor comparison, but it is imprecise and the built-versus-cancelled design does not confirm it.

The fiscal results are more selective than the original pooled specifications suggested. Table 2 reports the same not-yet-treated estimator and state-cluster bootstrap used in the county audits. Local school property-tax revenue rises by 16.6% ($p = 0.07$) with no detectable pretrend. Capital outlay per pupil rises by 28.5%, but is imprecisely estimated ($p = 0.17$). Total local revenue and local revenue per pupil do not rise. The county highway estimate is positive and marginally significant, but moves materially when the date spine is updated. The broad public-works series fails its pretrend test. We therefore treat the school property-tax increase as suggestive and do not interpret the other fiscal estimates as robust effects.

Table 2: Public-finance effects after hyperscale openings

Outcome	Geography	Effect	Post p	Pretrend p
Local property-tax revenue	school district	+16.6%	0.07	0.32
Total local revenue	school district	−2.3%	0.83	0.56
Local revenue per pupil	school district	+2.0%	0.76	0.16
Capital outlay per pupil	school district	+28.5%	0.17	0.51
Highway current + construction	county, balanced	+38.9%	0.07	0.13
Broad public works	county, balanced	+9.3%	0.41	< 0.01

Notes: Cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated geographies. School data are NCES/Census F-33 finance files for 2012–2023; county data are Census local government finance files for 2012–2023. Outcomes are adjusted for state-by-year means before the cohort comparisons. Inference uses 499 state-cluster bootstrap draws. Pretrend p jointly tests event years -4 through -2. The highway estimate moves materially across operational-date versions and is treated as a sensitivity result rather than a headline effect.

Taken together, the non-labor outcomes do not point to a broad local demand boom. There is modest evidence of housing capitalization and a suggestive school property-tax receipts, but no net migration, permitting, foot-traffic, or spending response that would indicate a large expansion of the local economy.

7 Incidence on electricity rates

The most common objection in data-center siting debates is that a hyperscale facility's load will raise electricity bills for nearby households. Wholesale-market modeling finds that existing data centers have raised average wholesale prices by 3 to 5 percent [Taylor et al., 2026]. Whether residential customers in a host utility's territory pay more is a separate question. Data centers select utilities with cheap power and available capacity, so a comparison of treated and untreated utilities must account for the siting choice.

We use Energy Information Administration Form 861 to compute each utility's average residential price as revenue over megawatt-hours sold from 2012 through 2024. A utility-state is treated in the year the first hyperscale facility opens in a county it serves. The clean hyperscale arm contains 52 utility-states first treated between 2015 and 2021 and no colocation facility. We compare them with roughly 2,000 utility-states that host no facility in the registry and, separately, with 54 utilities serving counties where an announced hyperscale project was cancelled.

Figure 2 reports a cohort-by-event-time estimator using not-yet-treated controls. Against never-treated utilities, the mean effect over years zero through five is -1.0% ($p = 0.42$), with a joint pretrend p-value of 0.97. The balanced panel, which retains utility-states observed in at least twelve of the thirteen years, gives -1.5% ($p = 0.17$). Against near-miss utilities, the full sample estimate is -0.9% ($p = 0.37$), and the year-five estimate is -1.9% ($p = 0.27$). Colocation estimates are small and positive, with a mean effect of $+0.7\%$ ($p = 0.54$).

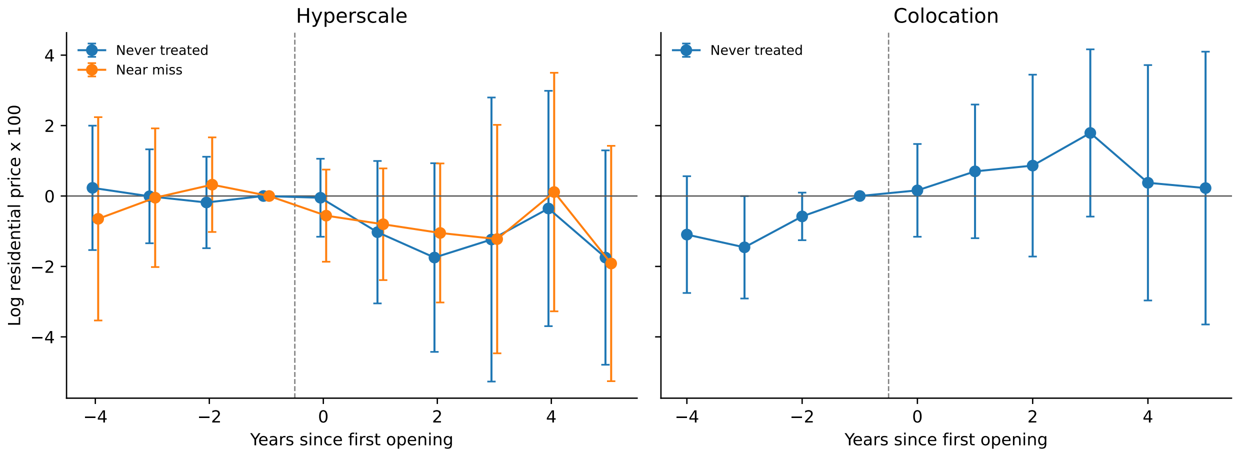

Figure 2: Residential electricity prices around the first data-center opening in a utility's territory
Notes: Cohort-by-event-time estimates of log average residential price (revenue over MWh, EIA Form 861 bundled-service utilities, 2012–2024). Hyperscale: 52 treated utility-states, 2,040 never-treated controls, and 54 near-miss controls. Colocation: 60 treated utility-states and the never-treated controls. The never-treated comparisons remove state-by-year means; the near-miss comparison removes year means. Inference uses 499 state-cluster bootstrap draws. Reference period is the year before opening; 95% confidence intervals shown.

The estimates reject the claim that host-utility residential rates rise sharply over this horizon. They are also consistent with a modest decline, especially in the balanced panel, but the full-sample and near-miss averages are imprecise. Spreading distribution-network fixed costs over a large new load is one possible explanation, as hyperscale facilities often use dedicated large-power tariffs. We do not observe the utility's underlying cost allocation or contract, however, so the design does not identify that mechanism. The results cover cloud-era cohorts through 2021 followed to 2024, not the larger post-2022 artificial-intelligence build-out.

8 Interpretation

The most stable result is the contrast between the physical scale of the investment and the limited response in nearby labor and firm outcomes. The capital footprint is large within the first kilometer and fades rapidly with distance. Advertised salaries and new-firm formation show no opening-date increase. Hiring and supplier postings are also centered near zero, but their confidence intervals admit moderate effects. The evidence therefore rejects a broad, precisely measured local boom; it does not establish that every possible spillover is exactly zero.

The pattern is consistent with the technology of a data center. A hyperscale campus employs dozens of workers rather than thousands. Its main inputs include electricity, network connectivity, construction services, and specialized equipment, much of which can be procured from regional or national suppliers. Its output is delivered to distant customers over the network. These features weaken the local labor-pooling, supplier, and customer-linkage channels that operate around a large manufacturing plant. The empirical results support this interpretation, but the supplier estimates are not precise enough to treat it as a complete decomposition.

The non-labor evidence is narrower. Rents are modestly higher in synthetic comparisons, net migration and multifamily permitting do not rise, and school property-tax receipts rise suggestively. Residential electricity rates show no increase and suggestive declines. These outcomes are consistent with some capitalization into land and local fiscal capacity, but the imprecise built-versus-cancelled rent estimate does not support a full welfare accounting.

Hyperscale and colocation facilities differ in both technology and location. Hyperscale campuses are generally larger and more likely to involve greenfield construction, while colocation facilities are multi-tenant and concentrated in denser markets. Their different outcome paths are informative, but they combine facility-type and market differences and should not be read as a randomized archetype comparison.

9 Conclusion

A hyperscale data center is as large a capital investment as the manufacturing plants that anchor the agglomeration literature, but its visible local response is much narrower. Relative to announced-then-cancelled sites, brightness at isolated built sites rises sharply when construction begins and fades to near zero within five kilometers. The result survives adjustment for the modest pre-construction slope, and daytime imagery shows vegetation clearing in the same period. We find no opening-date increase in advertised salaries or new-firm registration. Hiring and supplier-posting estimates are also centered near zero, although they are less precise. Workplace employment carries a pre-existing trend rather than a discrete opening effect. Detailed county-industry data cannot settle the downstream question cleanly because employment is heavily suppressed and several establishment series rise before opening.

The incidence evidence is more limited. Rents are modestly higher in synthetic comparisons, but the built-versus-cancelled estimate is imprecise. Multifamily permitting and net migration do not rise. School property-tax revenue rises suggestively, while capital outlays are positive but imprecise and county public-works estimates fail their pretrend tests. Residential electricity rates show no increase and suggestive declines. Foot traffic and commercial spending do not move robustly relative to cancelled sites.

The evidence therefore gives jurisdictions a narrower basis for evaluating place-based incentives than the large-plant analogy suggests. A hyperscale project clearly brings a large physical footprint. In the samples and horizons we study, however, we do not detect the accompanying labor-market and firm-entry responses that would support a broad local-multiplier claim. Some estimates, especially for hiring and suppliers, remain too imprecise to rule out moderate gains. Our estimates describe the cloud-era build-out through 2021 cohorts; the larger post-2022 artificial-intelligence campuses may have different effects and are not identified by this design.

A The instrument search

A natural way to address the siting-selection problem is an instrument for data center location. We searched for one and did not find a clean one. This appendix documents the search because the failure is informative. It is the reason we identify the effect from the within-site and county designs rather than from an instrument, and it characterizes how hyperscale siting actually works.

Predetermined siting instruments. We tested each plausibly exogenous, predetermined predictor of data-center siting, interacted where needed with the national capacity wave, and screened each against a manufacturing placebo (a valid instrument for data-center activity should not predict manufacturing employment). Table 3 summarizes the outcomes. Transmission-substation density predicts siting strongly but fails the placebo, because it is a proxy for industrial land. The 1980 urban college share, which works in the all-data-center county literature, carries the wrong sign for hyperscale, because hyperscale campuses go to low-college exurban land rather than to educated metros. Long-haul fiber proximity predicts siting only across metropolitan areas, not within a county, so it cannot identify a sub-county effect. Climate, measured by wet-bulb temperature and free-cooling days, does not predict siting at all, because the largest US hyperscale hub is hot and humid and operators engineer around climate to chase cheap power. Generation-capacity timing has no within-county predictive power. The pattern is systematic rather than bad luck. The features that predict hyperscale siting (power, industrial land, fiber) are the same features that are correlated with local industrial character, and the features that are clean (climate, historical human capital) do not predict where these facilities go. Hyperscale siting is cost-driven, and there is no clean predetermined instrument for it at sub-county scale.

Table 3: Predetermined siting instruments and why each fails

Instrument	First stage	Why it fails
345 kV substation density	strong	fails manufacturing placebo (industrial proxy)
1980 urban college share	moderate	wrong sign (hyperscale $\rightarrow$ low-college exurbs)
InterTubes fiber proximity	metro-scale	null within county
Climate (wet-bulb, free-cooling)	null	siting is cost-driven; the main hub is hot-humid
Generation-capacity timing	null	no within-county power

Notes: Each instrument is the predetermined county feature interacted with the national hyperscale capacity wave. "Fails placebo" means the instrument predicts manufacturing employment, violating the exclusion restriction.

A shift-share instrument is also weak. A shift-share design of the form used in concurrent county-level work [Alvarez et al., 2026], interacting the predetermined fiber and college exposures with national data-center demand growth, fares no better here, and the obstacle is the endogenous variable. Our public measures of data-center scale, namely capacity, square footage, and facility count, are lumpy and supply-driven, and a demand-side shift predicts a lumpy supply-side object poorly. The first stage with square footage as the endogenous reaches an F of only twenty to thirty, too weak to use.

Because the first stage is weak, we do not use its estimates as evidence. We note only, descriptively, that nothing in the instrumented results contradicts the paper's composition finding. The two-stage least-squares point estimates load on the direct data-center sector and are null or negative in the downstream industries. The instrument also loads on population-serving sectors such as health, the general area-development confound that the within-site and cancelled designs avoid, which is a further reason we treat the exercise as documentation of the search rather than as a design.

B The geography of compute

A common rationale behind data center incentives is that the facility will anchor a local digital economy. If the cloud servers are here, the reasoning goes, the firms that use the cloud will follow. This section examines that premise directly by mapping where the heavy users of compute actually sit relative to data centers, and by asking whether their location reflects the data center or something the data center does not provide. We use the Veridion firmographic database [Veridion, 2026], which geolocates 6.5 million US establishments and classifies each by industry, employment, and founding year. We define compute users by primary industry. Software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services form the compute-native core, with finance, scientific R&D, and semiconductors as a broader set.

B.1 Compute users cluster near data centers

Compute users are closer to data centers than the average firm. The median compute-native firm sits 11 kilometers from the nearest data center, against 20 kilometers for non-compute firms, and 69 percent of compute users fall within 25 kilometers of a data center, against 55 percent of other firms (Table 4). The gradient runs as one would expect from the technology. Cloud and hosting firms are nearest at a 6 kilometer median, software firms next at 9 kilometers, while telecommunications and streaming, whose footprints are more dispersed, sit farthest out.

This proximity is not an artifact of headquarters location. Measuring each firm's operating footprint by the geography of its job postings rather than its headquarters returns the same pattern at a slightly larger distance, because a firm's branch offices spread beyond its headquarters metro. Nor is it merely that both compute users and data centers crowd into a handful of tech metros. Removing the 45 largest metropolitan areas, compute users remain closer to the surviving data centers than non-compute firms are (a 75 versus 87 kilometer median, with software firms among the closest), so the association holds in second-tier and rural America as well.

Table 4: Distance from firms to the nearest data center

	Firms	Median km	≤ 25 km	≤ 50 km
Panel A: Headquarters location (Veridion)				
Non-compute firms	5,692,641	20.5	55%	70%
Compute users (core)	309,760	11.0	69%	81%
Cloud and hosting	26,188	6.1	75%	84%
Software	101,757	8.9	72%	84%
Computer systems design	154,934	11.7	68%	81%
Finance	193,540	15.9	62%	75%
Telecommunications	10,125	22.2	53%	66%
Panel B: Establishment footprint (job postings)				
Compute users (core)		15.1	63%	76%
Panel C: Outside the 45 largest metros				
Non-compute firms	2,603,195	86.8	20%	33%
Compute users (core)	103,356	75.3	30%	41%

Notes: Distance from each firm to the nearest hyperscale-clean data center, by haversine. Panel A places each firm at its Veridion headquarters coordinate. Panel B replaces the headquarters with the firm's job-posting locations (WageScape), weighted by the number of distinct hiring firms at each location, so it reflects the operating footprint rather than the headquarters. Panel C excludes all firms and data centers within 50 km of the 45 largest metropolitan and technology-hub centers and measures distance to the nearest remaining data center. Compute-native "core" sectors are software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services.

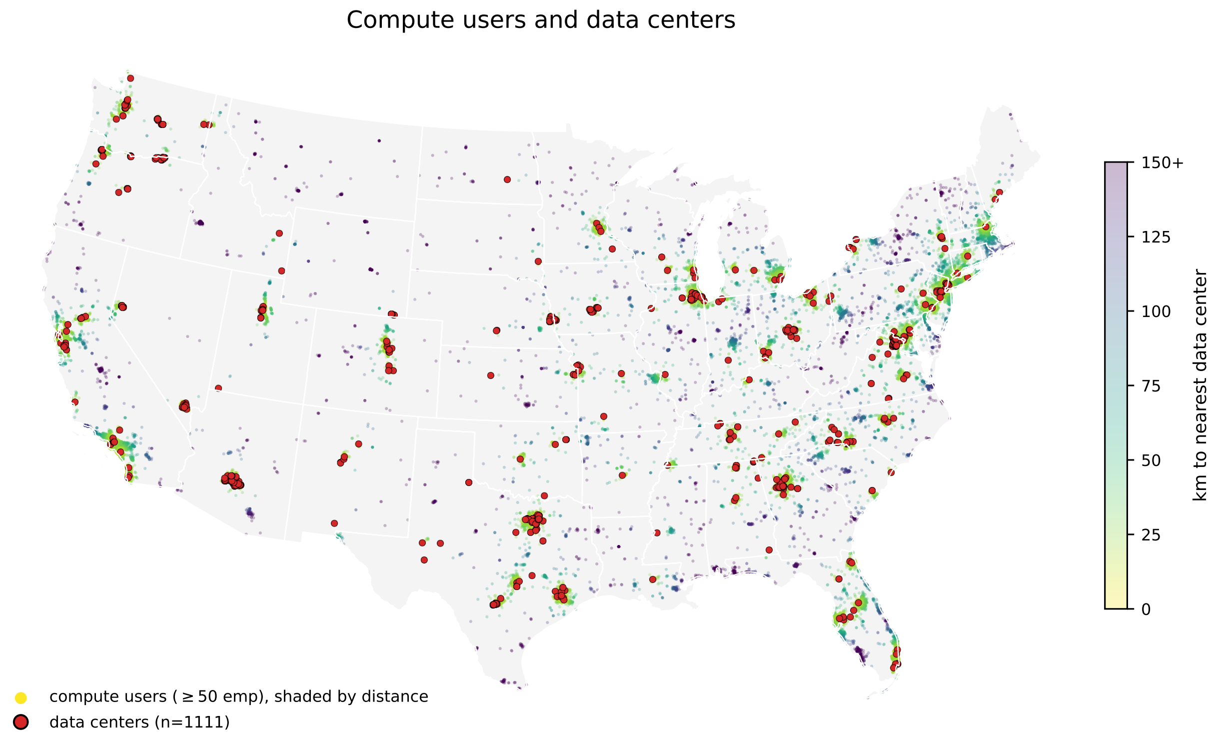

Figure 3: Compute-user firms and data centers across the continental United States. Points are compute-native firms with at least 50 employees, shaded by distance to the nearest data center, with data centers marked in red. Both concentrate in the same coastal and metropolitan corridors, while interior firms shade dark, far from any facility.

The two populations trace the same map (Figure 3), crowding into the coastal and metropolitan corridors and thinning together across the interior. The pull is strongest for the largest firms. When sorted by employment, the median distance to a data center falls monotonically from 13 kilometers for the smallest firms to 4 kilometers for those with more than ten thousand employees, and the dispersion collapses with it. The biggest compute users cluster tightly around data centers, and almost none are far from one. Only about 5 percent of compute firms with at least a thousand employees sit more than 50 kilometers from any data center. The handful that do are software companies that scaled in place in second-tier cities, such as Esri in Redlands, Jack Henry in rural Missouri, and Ansys outside Pittsburgh. Distance to compute is a constraint these firms evidently did not face, because the cloud let them consume capacity hundreds of kilometers away.

B.2 Two kinds of data center hub

The co-location is a metropolitan phenomenon, and seeing this matters for what follows. Grouping data centers into hubs reveals a sharp split. Metropolitan hubs are embedded in dense clusters of sector-matched compute users. The 239 data centers around Ashburn, Virginia sit among thousands of government IT contractors, with DXC, Leidos, Booz Allen, SAIC, and CACI all within 25 kilometers. The Bay Area hubs are ringed by Google, Broadcom, Synopsys, and ServiceNow. By contrast, the large exurban hyperscale clusters are isolated. The 50 data centers around Hillsboro, Oregon and the cluster near Quincy, Washington each have on the order of four compute firms within 25 kilometers. The same physical infrastructure produces a thick compute neighborhood in a tech metro and an empty one in the exurbs.

B.3 Why they co-locate, and why it is not the data center

Part of the explanation is shared infrastructure. Data centers and compute users both require access to the high-voltage grid, and both sit closer to it than ordinary firms. The median data center is 3.4 kilometers from a 230-kilovolt substation, the median compute user 5.0 kilometers, and the median other firm 6.4 kilometers. High-voltage substation density independently predicts where compute users locate. But the grid is only part of the story. The proximity between compute users and data centers survives controlling for substation access, so a common pull toward the power backbone, together with the fiber and talent that concentrate in metros, leaves the two in the same places, consistent with shared infrastructure rather than one drawing the other.

Three pieces of evidence indicate that the data center is not the cause. First, the compute users near data centers predate them. Among compute firms within 25 kilometers of a data center, at least 69 percent were founded before the nearest data center opened. Because the firm registry over-represents recently founded firms, the true share is higher. The local compute base existed first. Ashburn illustrates the direction of causation. The government technology ecosystem of Northern Virginia and its early internet exchange drew the data centers to the region, not the reverse. Second, the timing evidence in Section 3.5 shows that data center openings do not raise job postings in the surrounding rings, for compute-using industries or any other, once a smooth pre-existing trend is removed. The opening is not a turning point for local hiring. Third, and most directly, exurban data centers sit in compute deserts and stay there. The median metropolitan data center has roughly 2,700 compute firms within 25 kilometers, while the median exurban data center has 41 (Table 5). The facility does not move that number.

Table 5: Compute users near metropolitan versus exurban data centers

	Data centers	Median compute firms ≤ 25 km	Share with < 5 nearby
Metropolitan data centers	824	2,702	0%
Exurban data centers	287	41	11%

Notes: Each data center is classified as metropolitan if it lies within 50 km of one of the 45 largest metropolitan or technology-hub centers, and exurban otherwise. The middle column reports the median count of compute-native firms (Veridion) within 25 km of the data center; the last column the share of data centers with fewer than five compute firms within 25 km. Separately, among compute firms within 25 km of any data center, 69 percent were founded before the nearest data center opened.

B.4 Implication for place-based policy

The policy reading is direct. A jurisdiction weighing a data center, especially an exurban one, should not expect compute users to arrive in its wake. Firms that consume cloud capacity locate near transmission, fiber, and skilled labor. These assets concentrate in metropolitan areas, and a single data center does not create them. The compute users already near data centers were mostly there first, and the data centers that open in compute-sparse places remain in compute-sparse places. What the facility delivers locally is the capital footprint and the rent pressure documented above, not a cluster of digital firms. The hope that the servers will bring the software is, on this evidence, misplaced.

C Supplementary results

Table 6 reports every outcome under every design in which it can be estimated, so the routing rule of Section 3 can be audited cell by cell rather than taken on faith.

Table 6: Every outcome under every design: estimates and routing

Outcome (hyperscale, inner ring)	Built vs. cancelled	Ring / synthetic	County evidence
Capital (nighttime lights)	**+70% ($p < 0.01$)**	**+38% ($p < 0.01$)**	—[a]
Residential rent	+4.3% ($p = 0.23$)	**+2 to +5%**[b]	—[a]
Advertised salary	−11.6% ($p = 0.54$)	**−0.9% ($p = 0.77$)**	—[a]
Job postings	**+4.0% ($p = 0.75$)**[c]	excluded[d]	—[a]
Foot traffic	**−12.0% ($p = 0.12$)**[e]	excluded[d]	—[a]
Consumer spending	**+4.6% ($p = 0.60$)**	+8.5% ($p = 0.50$)[f]	—[a]
Firm entry (SoS registrations)	—[g]	**−0.5% ($p = 0.62$)**[h]	+1.4% ($p = 0.49$)[i]
Workplace employment	—[g]	excluded[d]	inconclusive[j]

Notes: Post-event estimates for the hyperscale arm at the innermost ring each design supports (0–1 km for nighttime lights, 0–5 km otherwise). Bold marks the design that carries the paper's interpretation for that outcome. Built-versus-cancelled capital estimate is the isolated-sample groundbreaking-year jump; magnitudes at longer horizons rest on few cancelled controls (Section 4).

[a] Not estimated at the county level; no county analog of the ring outcome.

[b] Synthetic-control headline; the raw within-site cells (+6.6 to +9.4%) are an upper bound (Section 6).

[c] The v9 analysis-date spine estimate is imprecise. Alternative count, long-difference, and construction-dated specifications likewise do not identify a precise hiring response. Minimum detectable increases remain large.

[d] Fails the flat-leads gate: a pre-existing inner-versus-outer trend the ring design cannot separate from an effect (Section 3).

[e] Falls to −3.0% ($p = 0.68$) on the balanced panel, so not read as a reliable displacement.

[f] Reported with caveats in Section 6; passes a random-donor placebo.

[g] No cancelled-arm panel exists for this outcome (business registrations and LODES cannot be reconstructed around never-built sites at ring scale).

[h] Hyperscale-only sample under the v9 analysis-date spine; the pooled all-archetype estimate is +0.7% ($p = 0.24$).

[i] Census Business Formation Statistics, national county design.

[j] Detailed QCEW employment is heavily suppressed, while establishment counts in several sectors rise before opening. The disclosure-corrected, not-yet-treated estimates therefore do not identify either a downstream gain or a precise zero (Section 5.5).

Data Availability Statement

This study combines publicly available data with several proprietary datasets obtained under license. All code that constructs the analysis datasets from the raw sources and reproduces every table and figure will be deposited in the AEA Data and Code Repository. The deposit includes the publicly available inputs and all author-constructed files, namely the data-center registry, the operating and construction-start dates, and the announced-then-cancelled cohort. The proprietary inputs cannot be redistributed. For each, we report the provider and access terms and include the exact product identifiers, date ranges, geographic filters, and extraction code, so that a licensed user can regenerate the facility-by-ring panels and reproduce all results.

Publicly available data.

- Satellite nighttime lights: VIIRS Day/Night Band Monthly Composite (NOAA, product VCMCFG), accessed through Google Earth Engine.
- Daytime imagery for construction dating: Sentinel-2 and Landsat surface reflectance (ESA and USGS), accessed through Google Earth Engine.
- Data-center registry: base layer from the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), augmented by the authors from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, trade press, DatacenterMap, Cloudscene, and data-center REIT filings. The compiled registry, dates, and cancelled cohort are in the deposit.
- Construction timing: Federal Aviation Administration Obstruction Evaluation / Airport Airspace Analysis filings (Form 7460-1).
- Employment and establishments: Quarterly Census of Employment and Wages (Bureau of Labor Statistics) and County Business Patterns (Census Bureau).
- Migration and income: IRS Statistics of Income county-to-county migration data.
- Housing supply: Census Building Permits Survey. Population: 2020 Census Centers of Population (tract population-weighted centroids).
- Workplace employment: LEHD Origin-Destination Employment Statistics (LODES, Census Bureau).
- Business formation: Census Business Formation Statistics, and state business-registration bulk files for Virginia, Texas, and Iowa (Secretaries of State and data.iowa.gov).
- Local public finance: Census of Governments and the Annual Survey of School System Finances (F-33).
- Electricity: U.S. Energy Information Administration Form 861.

- Instrument-search inputs: the InterTubes long-haul fiber map [Durairajan et al., 2015]; the 1980 county urban-college share (IPUMS NHGIS); and transmission-substation locations (Homeland Infrastructure Foundation-Level Data).

Proprietary data (available under license, not redistributable).

- Job postings: LinkUp Job Records [LinkUp, 2025]; WageScape salary postings [WageScape, 2022].
- Residential rents: RentHub Rental Data [RentHub, 2022].
- Consumer spending: SafeGraph Spend Patterns [SafeGraph, 2022].
- Foot traffic: Advan Weekly Patterns [Advan Research, 2025].
- Firmographics: Veridion firm master data.

The five Dewey-distributed datasets are available to researchers through a subscription to Dewey Data (`https://www.deweydata.io`); Veridion firmographics are available under license from Veridion (`https://veridion.com`). The authors accessed these data under institutional license and are not permitted to redistribute the raw files. The deposit provides the dataset identifiers and the extraction and aggregation code required to rebuild the analysis files from a licensed copy.

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